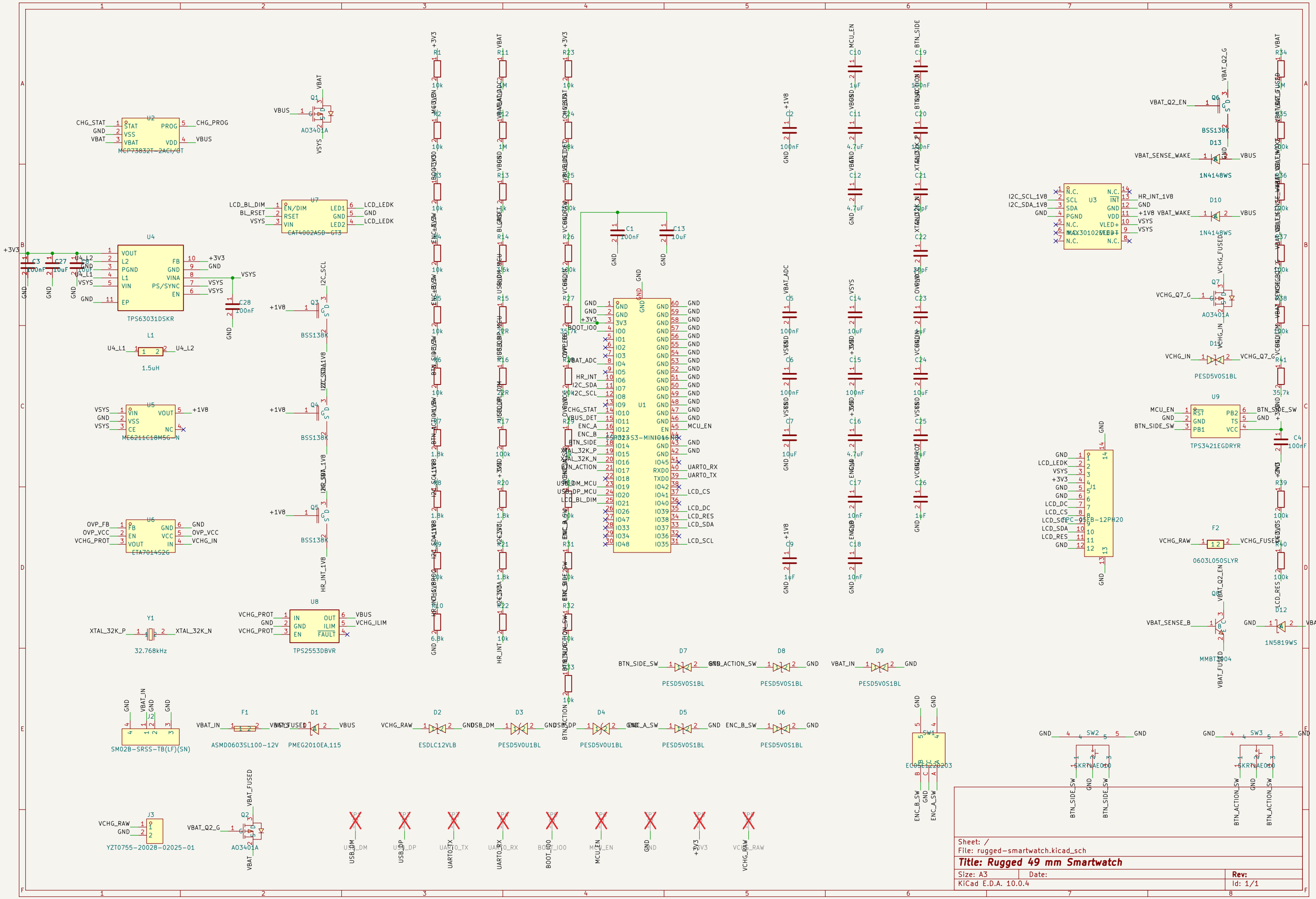

U2
MCP73832T-2ACI/OT
CHG_STAT
GND
VBAT
STAT
VSS
VBAT
PROG
VDD
CHG_PROG
VBUS
Q1
AO3401A
VBAT
VSYS
U7
CAT4002ASD-GT3
LCD_BL_DIM
BL_RSET
VSYS
EN/DIM
RSET
VIN
LED1
GND
LED2
LCD_LEDK
U4
TPS63031DSKR
+3V3
VOUT
L2
PGND
L1
VIN
FB
GND
VINA
PS/SYNC
EP
EN
VSYS
C3
C27
C29
C28
100nF
10uF
L1
U4_L1
U4_L2
1.5uH
U5
ME6211C18M5G-N
VSYS
VIN
VSS
CE
VOUT
NC
+1V8
U6
ETA7014S2G
OVP_FB
OVP_VCC
VCHG_PROT
FB
EN
VOUT
GND
VCC
IN
OVP_VCC
VCHG_IN
Y1
32.768kHz
XTAL_32K_P
XTAL_32K_N
U8
TPS2553DBVR
VCHG_PROT
GND
IN
EN
OUT
ILIM
FAULT
VBUS
VCHG_ILIM
Q3
Q4
Q5
BSS138K
I2C_SCL
I2C_SDA
HR_INT_1V8
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
10k
1.8k
6.8k
R11
R12
R13
R14
R15
R16
R17
R18
R19
R20
R21
R22
100k
1M
C1
C13
100nF
10uF
U1
ESP32S3-MINI-1-N8
3V3
BOOT_IO0
VBAT_ADC
HR_INT
I2C_SDA
I2C_SCL
CHG_STAT
VBUS_DET
ENC_A
ENC_B
BTN_SIDE
XTAL_32K_P
XTAL_32K_N
BTN_ACTION
USB_DM_MCU
USB_DP_MCU
LCD_BL_DIM
MCU_EN
UART0_RX
UART0_TX
LCD_CS
LCD_DC
LCD_RES
LCD_SDA
LCD_SCL
R23
R24
R25
R26
R27
R28
R29
R30
R31
R32
R33
47k
C2
C5
C6
C7
C9
100nF
10uF
1uF
C10
C11
C12
C14
C15
C16
C17
C18
C19
C20
C21
C22
C23
C24
C25
C26
1uF
4.7uF
10nF
20pF
U3
MAX30102EFD+T
N.C.
SCL
SDA
PGND
VLED+
INT
GND
VDD
I2C_SCL_1V8
I2C_SDA_1V8
HR_INT_1V8
VBAT_SENSE_WAKE
Q6
BSS138K
VBAT_Q2_G
VBAT_Q2_EN
D13
1N4148WS
D10
VBUS
VCHG_FUSED
Q7
AO3401A
VCHG_Q7_G
D11
PESD5V0S1BL
VCHG_IN
U9
TPS3421EGDRYR
MCU_EN
BTN_SIDE_SW
RST
GND
PB1
PB2
TS
VCC
BTN_SIDE
J1
FPC-05B-12PH20
LCD_LEDK
VSYS
+3V3
LCD_DC
LCD_CS
LCD_SCL
LCD_SDA
LCD_RES
R34
R35
R36
R37
R38
R39
R40
R41
10k
35.7k
100k
F2
0603L050SLYR
VCHG_RAW
VCHG_FUSED
Q8
MMBT3904
VBAT_SENSE_B
VBAT_FUSED
D12
1N5819WS
C4
100nF
J2
SM02B-SRSS-TB(LF)(SN)
VBAT_IN
F1
ASMD0603SL100-12V
D1
PMEG2010EA,115
D2
ESDLC12VLB
D3
PESD5V0U1BL
D4
D5
D6
PESD5V0S1BL
D7
D8
D9
SW1
EC05E1220203
ENC_B_SW
ENC_A_SW
SW2
SW3
SKRTLAE010
BTN_SIDE_SW
BTN_ACTION_SW
J3
YZT0755-20028-02025-01
VCHG_RAW
VBAT_Q2_G
Q2
AO3401A
USB_DM
USB_DP
UART0_TX
UART0_RX
BOOT_IO0
MCU_EN
GND
+3V3
VCHG_RAW
Sheet: /
File: rugged-smartwatch.kicad_sch
Title: Rugged 49 mm Smartwatch
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1